CBSE Test Paper 04

Chapter 12 Thermodynamics

1. What amount of heat must be supplied to 2.0 $\times$ 10^{-2} kg of nitrogen (at room temperature) to raise its temperature by 45 $^\circ$C at constant pressure? (Molecular mass of N_2 = 28; R = 8.3 J mol^{-1} K^{-1}.) **1**
   a. 904 J
   b. 974 J
   c. 954 J
   d. 934 J
2. Two cylinders A and B of equal capacity are connected to each other via a stopcock. A contains a gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stopcock is suddenly opened. Do the intermediate states of the system (before settling to the final equilibrium state) lie on its P-V-T surface? **1**
   a. yes
   b. no for points on the left
   c. No
   d. yes for points on the left
3. The spontaneous processes of nature are **1**
   a. irreversible
   b. static
   c. reversible
   d. dynamic
4. The efficiency (η) of a heat engine is defined by (W is the output and Q_1 the heat input) **1**
   a. $\eta = \frac{Q_1}{W}$
   b. $\eta = \frac{2W}{Q_1}$

c. $\eta = \frac{W}{2Q_1}$
d. $\eta = \frac{W}{Q_1}$

5. A 1.0-mol sample of an ideal gas is kept at 0.0°C during an expansion from 3.0 L to 10.0 L. If the gas is returned to the original volume by means of an isobaric process, how much work is done by the gas? **1**

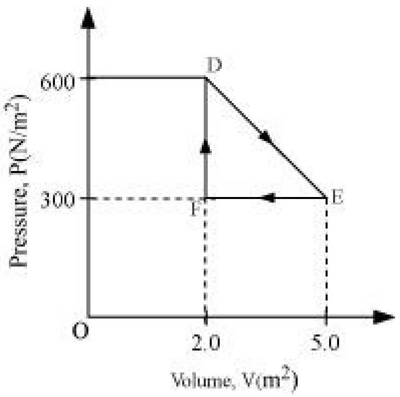
   a. -1.6 $\times$ 10^3 J
   b. -1.4 $\times$ 10^3
   c. -1.3 $\times$ 10^3
   d. -1.5 $\times$ 10^3

6. Write conditions for an isothermal process. **1**

7. On a winter night, you feel warmer when clouds cover the sky than when sky is clear. Why? **1**

8. Under what condition, an ideal Carnot engine has 100% efficiency? **1**

9. A thermodynamic system is taken from an original state to an intermediate state by the linear process shown in figure. **2**



   Its volume is then reduced to the original value from E to F by an isobaric process. Calculate the total work done by the gas from D to E to F.

10. If a body is heated from 27^0 C to 927^0 C then what will be the ratio of energies of radiation emitted? **2**

11. Find molar specific heat for the process $p = \frac{a}{T}$ for a monoatomic gas. **2**

12. A brass boiler has a base area of 0.15 m^2 and thickness 1.0 cm. It boils water at the

rate of 6kg/min when placed on a gas stove, Estimate the temperature of the part of flame in contact with the boiler. Thermal conductivity of brass = 109J/s/m/^{0}C, heat of vaporization of water = 2256 J/g? **3**

13. Temperatures of the hot and cold reservoirs of a Carnot engine is raised by equal amounts. How the efficiency of the Carnot engine affected? **3**

14. A gaseous mixture enclosed in a vessel consists of 1 g mole of a gas A$\left(\gamma = \frac{5}{3}\right)$ and some amount of gas B$\left(\gamma = \frac{7}{5}\right)$ at a temperature T. The gases A and B do not react with each other and are assumed to be ideal.
Find the number of gram moles of the gas B, if γ for the gaseous mixture is $\left(\frac{19}{13}\right)$. **3**

15. Consider that an ideal gas (*n* moles) is expanding in a process given by *P* = f (V), which passes through a point (V_0, P_0). Show that the gas is absorbing heat at (P_0, V_0) if the slope of the curve P = f (V) is larger than the slope of the adiabat passing through (P_0, V_0). **5**